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Introduction, Standard test signals, Unit step response of First and Second order Systems. Transient specifications of second order systems, steady state errors

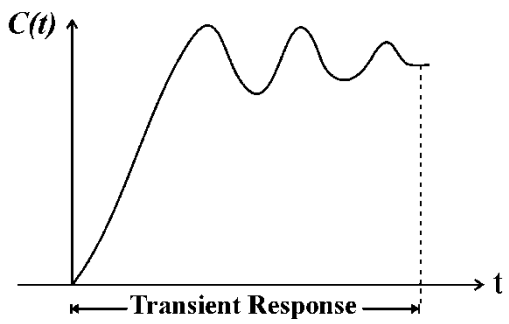
Unit - 4

Time Response of feedback control systems

4.1 Introduction

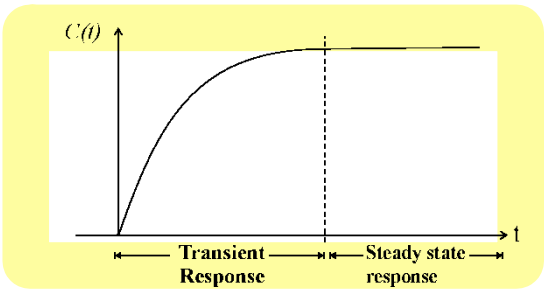
In time domain analysis, time is the independent variable when a system is given an excitation, there is a response (output). This response varies with time and is called the time response. Time domain analysis means the study of output or response of the system as function of time for a given input signal. The time domain analysis can be classified into two parts that is transient response and steady state response.

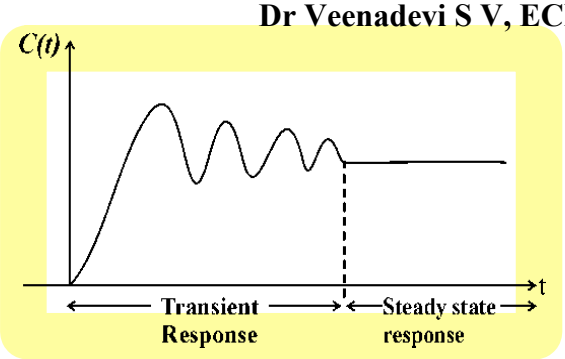
Transient Response is defined as that part of the response which goes to zero as time 't' reaches infinity. It remains only some time from initial state to final state.



$$\lim_{t\to\infty} C_t(t) = 0$$

Steady state response is defined as the response of the system as time tends to infinity.





$$Css(t) = \lim_{t \to \infty} C(t)$$

The system shows transient response because of the presence of energy storing elements such as mass, capacitance or inductance in the system. In the absence of energy storing elements the system response consists only of steady state portion.

In many control systems the input to the system are not known in advance. For example: In radar tracking systems the position of the target is one of the inputs to the system. But this input signal changes in random manner. Therefore there is a need for some standard test input signals. Thus the system cannot be done from input point of view as we are unable to know in advance the type of input.

The test signal to be used for the purpose depends on the nature of input to which the system is subjected to during its operation in the field.

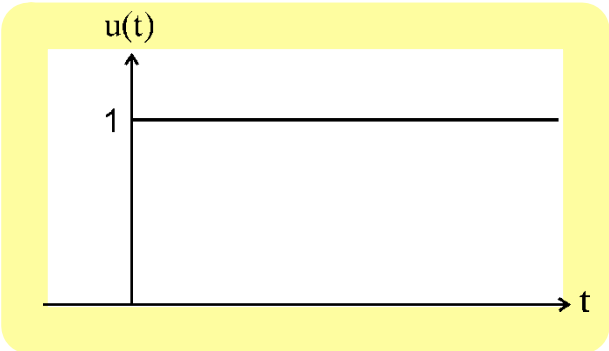
4.2 Standard test input signals

Unit Step input (Step displacement input)

This is the sudden change in the reference input signal. Mathematically represented as

$u(t) = 1 \quad t > 0$

$\quad\quad = 0 \quad t < 0$



If $r(t) = A\,u(t)$, where A is a constant.

i.e. $u(t) = A \quad t > 0$

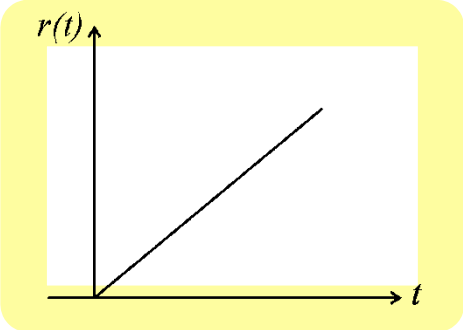
$\quad\quad\quad = 0 \quad t < 0$

Laplace transform of u(t) i.e $L[u(t)] = A.\frac{1}{s}$

Unit Ramp function (Step Velocity input)

If the reference input signal is changing gradually at a constant rate then such a signal is represented by a Ramp function.

Mathematically represented by

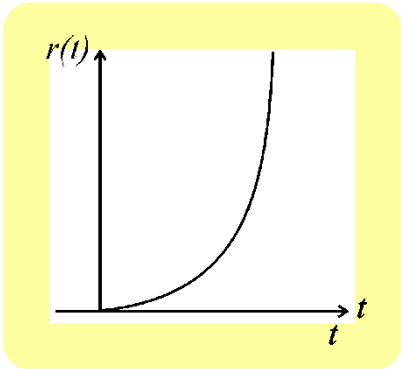


$$r(t) = A\,t \qquad t > 0$$
$$= 0 \qquad t < 0$$

Or $r(t) = A\,.\,t\,u(t)$

Laplace transform $L[r(t)] = A\,\frac{1}{s^2}$

Unit Parabolic function (Acceleration input)

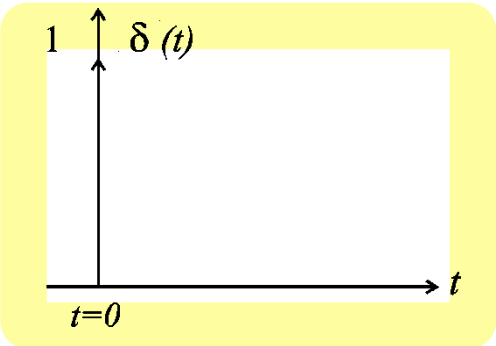


$$r(t) = \frac{A}{2}t^2 \qquad t > 0$$
$$= 0 \qquad t < 0$$

Or $r(t) = \frac{A}{2}t^2\,u(t)$

$L\,[r(t)] = L[\frac{t^2}{2}] = \frac{1}{S^3}$

Impulse input: If the reference input signal of large amplitude is applied for very short duration, then such an input is represented by an impulse function.



$$\delta(t) = 1, \quad t = 0$$
$$= 0 \quad t \neq 0$$

and $\int_{-\infty}^{\infty} \delta(t)\,.\,dt = 1$

Then $LT[\delta(t)] = 1$

From the transient response, we can know

(i) When the system begins to respond after an input is given.

(ii) How much time it takes to reach the output for the first time.

(iii) Whether the output oscillates about its final value.

(iv) When does it settle to the final value.

The above information is translated to 'performance index' and will form an important requirement in design.

Steady State response

(i) How long it took before steady state was reached.

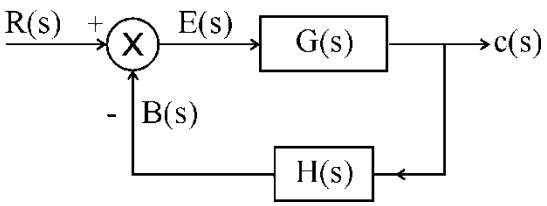
(ii) Whether there is any error between the desired and actual values.

(iii) Whether this error is constant, zero or infinite.

(iv) t judge the accuracy of the system output.

Derivation of Steady State error

Steady State response is important to judge the accuracy of the output.



From the above figure.

Steady State error $E(S) = R(s) - B(s)$

$B(s) = C(s) \cdot H(s)$

$\therefore \quad E(s) = R(s) - C(s)\,H(s)$

Also $C(s) = G(s)\,E(s)$

$\therefore \quad E(s) = R(s) - G(s)\,E(s)\,H(s)$

$\therefore \quad E(s)\,[1 + G(s)\,H(s)] = R(s)$

$$E(s) = \frac{R(s)}{1+G(s)H(s)}$$

If positive feedback is used then,

$$E(s) = \frac{R(s)}{1-G(s)\,.\,H(s)}$$

$$\therefore \quad E(s) = \frac{R(s)}{1\pm G(s)\,.\,H(s)}$$

For a unity feedback system H(s) = 1

$\therefore$ $E(s) = \frac{R(s)}{1\pm G(s)}$ In time domain e(t) $= L^{-1}\,[E(s)]$

$\therefore$ This is the expression of errors valid for all time. Steady state error is defined as

$$e_{ss} = \underset{t\to\infty}{lt}\, e(t)$$

$$= \underset{s\to 0}{lt}\, S.E(s)$$

$\because$ From the final value theorem in LT.

$$e_{ss} = \underset{s\to 0}{lt}\, S\,.\,\frac{R(s)}{1+G(s).H(s)}$$

$$x(\infty) = \lim_{s\to 0} S.X(s)$$

Unity gain system , e(t) = r(t) – c(t)

Type of a system : The number of poles of a system having open loop transfer function G(s) H(s) specifies the type of the system.

4.3a) Unit step response of first order system

A system is said to be of first order if the highest power of 'S' in the denominator polynomial of the transfer function is unity.

Consider the I (first) order system shown in figure.

$$R(S) \rightarrow \boxed{\frac{1}{1+TS}} \rightarrow C(s)$$

The transfer function of the system is $G(s) = \frac{C(s)}{R(s)} = \frac{1}{Ts+1}$

The transfer function of the system is $M(s) = \frac{C(S)}{R(s)} = \frac{1}{1+TS}$

For the unit step input r(t) = u(t)

$R(s) = LT[r(t)] = \frac{1}{S}$

The output response is given by $C(s) = R(s).M(s) = \frac{1}{S(Ts+1)}$

On partial fraction expansion,

$C(s) = \frac{1}{S(Ts+1)} = \frac{A}{S} + \frac{B}{Ts+1}$

$1 = A\,(Ts + 1) + BS$

$S = 0,\ 1 = A$

$S = -\frac{1}{T},\ \ 1 = A\,(T\left(-\frac{1}{T}\right) + 1) + B\left(-\frac{1}{T}\right)$

$1 = -\frac{1}{T}\,.\,B$

$B = -\,T$

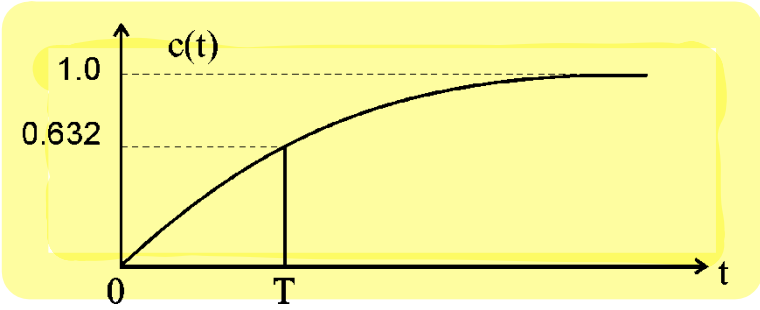
$C(s) = \frac{1}{S} - \frac{T}{Ts+1}$

Taking inverse LT. $\Longrightarrow$ $C(t) = 1 - e^{-t/T}$, $t \geq 0$

Let us plot some values of C(t)

t	C(t)
T	$C(t) = 1 - e^{-1} = 0.632$
2T	0.86
3T	0.95
4T	0.982
5T	0.993
∞	1

Here the output C(t) is initially zero and exponentially reaches a final value of unity as shown in figure.



An important characteristics of the response is that at t = T, C(t) = 0.632 or the response reaches 63.2% of the final value. Here time constant T is the time required for the output to reach 63.2% of its final value.

If a system has a small time constant then it has fast response.

The initial slope of the curve at t = 0 is given by

$$\frac{dc(t)}{dt} = \frac{1}{T} e^{-t/T} \Big|_{t=0} = \frac{1}{T}$$

Steady State error

$$e(t) = r(t) - C(t)$$

$$= 1 - (1 - e^{-t/T})$$

$$= 1 - 1 + e^{-t/T}$$

$$e(t) = e^{-t/T}$$

$$e_{ss} = \underset{t\to\infty}{lt}\, e(t) = \underset{t\to\infty}{lt}\, e^{-t/T} = 0$$

Note :

(i) Steady state error is zero.

(ii) First order system is capable of responding to a unit step unit without any steady state error.

(iii) The speed of response depends on constant [T], smaller the time constant, larger the response.

Time response of first order system for unit impulse input:

$C(s) = \frac{1}{1+Ts} R(s)$

$r(t) = \delta(t)$

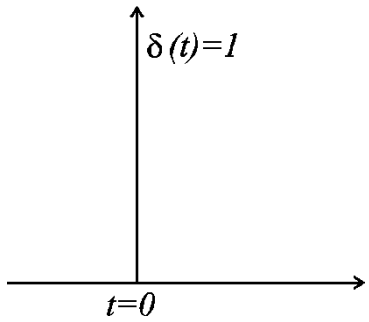
$R(s) = 1$

$C(s) = \frac{1}{Ts+1} \cdot 1 = \frac{(\frac{1}{T})}{(S+1/T)}$ (1)

Taking laplace inverse $C(t) = \frac{1}{T} \cdot e^{-t/T}$

$e(t) = r(t) - C(t)$

$= \delta(t) - \frac{1}{T} e^{-t/T}$



$e_{ss} = \underset{t\to\infty}{lt} [\delta(t) - \frac{1}{T} e^{-t/T})$

$= \delta(\infty) - \frac{1}{T} e^{\infty}$

$= 0 - 0$

$e_{ss} = 0$

Time response of first order system for unit Ramp input:

$C(s) = \frac{1}{1+Ts} \cdot R(s)$

$r(t) = t \cdot u(t)$

$R(s) = \frac{1}{S^2}$

$C(s) = \frac{1}{1+Ts} \cdot \frac{1}{S^2} = \left\{\frac{\frac{1}{T}}{s+\frac{1}{T}}\right\} \cdot \frac{1}{S^2}$

$= \frac{A}{S} + \frac{B}{S^2} + \frac{C}{s+\frac{1}{T}}$

$\therefore B = C(S) \cdot S^2 |_{s^2=0} = \frac{\left(\frac{1}{T}\right)}{\left((S+\frac{1}{T}\right)} \cdot \frac{S^2}{S^2} |_{S=0} = 1$

$\therefore$ B = 1

$$C = C(S) \cdot \left(S + \frac{1}{T}\right) \Big|_{s = -\frac{1}{T}}$$

$$= \frac{\frac{1}{T}}{(s + \frac{1}{T})} \cdot \left(S + \frac{1}{T}\right) \Big|_{s = -\frac{1}{T}}$$

C = T

A = - T, B = 1, C = T

$$\therefore\ C(S) = \frac{-T}{S} + \frac{1}{S^2} + \frac{T}{S + \frac{1}{T}}$$

Taking inverse L.T.

$$C(t) = -T + t.\,u(t) + T \cdot e^{-t/T}$$

$$= T\left(e^{-t/T} - 1\right) + t$$

$$= t - T\left(1 - e^{-t/T}\right)$$

The error signal

$$e(t) = r(t) - C(t)$$

$$= t - \left\{t - T(1 - e^{-t/T})\right\}$$

$$= T\left(1 - e^{-t/T}\right)$$

Steady state error

$$ess = \lim_{t \to \infty} e(t) = \lim_{t \to \infty} T\left(1 - e^{-t/T}\right)$$

$$= T - e^{\infty} = T - 0 = T$$

$\therefore$ ess = T

Standard Test Input Signals

Input Type	Description / Physical Meaning	Time Domain Expression	Laplace Transform	Purpose / Use
Unit Step Input	Sudden change in input magnitude	$u(t) = 1,\ t > 0$	$\frac{1}{s}$	Used to study transient and steady-state behavior
Unit Ramp Input	Input increases linearly with time	$r(t) = t$	$\frac{1}{s^2}$	Represents constant velocity systems
Unit Parabolic Input	Input increases quadratically with time	$r(t) = \frac{t^2}{2}$	$\frac{1}{s^3}$	Represents constant acceleration systems
Impulse Input	Very large input for a very short duration	$\delta(t)$	1	Used to study instantaneous system response

The first order system has a steady state error of T, when it is subjected to unit ramp input.

$$C(t) = t - T\left(1 - e^{-t/T}\right)$$

If T is Small -t/T increases $e^{-t/T}$ increases

e^{-St} is Small e^{-10t} *is large*

Smaller the time constant speed of response is more.

Note :

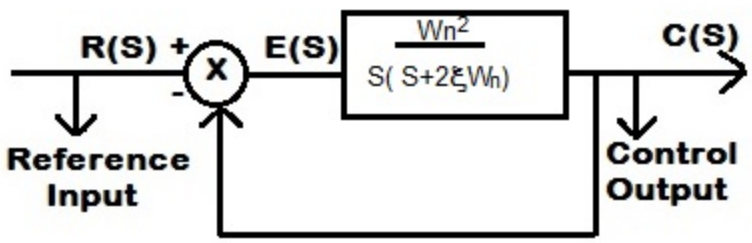
Unit step C(t) = 1 - $e^{-t/T}$, ess = 0

Unit impulse C(t) = $1/T\, e^{-t/T}$, ess = 0

Unit Ramp C(t) = t – T $\left(1 - e^{-t/T}\right)$, ess = T

4.3 b) Time response of second order system :

Consider the second order system as shown in figure.



A system is said to be of second order if the highest power of S in the denominator polynomial of the transfer function is two.

The overall transfer function of the system is

$$\frac{C(s)}{R(s)} = \frac{\frac{\omega_n^2}{S(S+2\,\zeta\,\omega_n)}}{\frac{1+\omega_n^2}{s(S+2\,\zeta\omega_n)}} = \frac{G(s)}{1+G(s)H(s)}$$

$$= \frac{\omega_n{}^2}{S(S+2\,\zeta\omega_n + \omega_n{}^2}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S(S+2\,\zeta\omega_n)+\,\omega_n{}^2}$$ is known as Standard form

Where ω_n = Natural undamped frequency.

ζ = damping ratio of the system.

$\sigma = \zeta\,\omega_n$ = damping factor.

$\omega_d = \omega_n\sqrt{1-\zeta^2}$ = damped frequency.

Characteristic equation :

$1 + G(s)\,H(s) = 0$

$\therefore\ S^2 + 2\,\zeta\omega_n\,s + \omega_n^2 = 0$

Poles of transfer function are same as the roots of characteristic equation.

$\zeta = 0$	undamped condition
$0 < \zeta < 1$	under damaped condition
$\zeta > 1$	over damaped condition
$\zeta = 1$	critically damped.

* For critically damped and over damaped systems the transient response of the system does not have oscillations.

* If $0 < \zeta < 1$, then the transient response due to unit step input is oscillatory and magnitude of oscillations become zero after sometime. The system is said to be underdamped.

* In undamped condition, the magnitude of oscillations in the transient response of the system does not becomes zero.

Time Response of underdamped system to unit step input $0<\zeta<1$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{S^2+2\,\omega_n\,\zeta\,S+\omega_n^2}$$

$$\therefore\ C(s) = \frac{\omega_n^2}{S^2+2\,\zeta\,\omega_n\,S+\omega_n^2}\ R(s)$$

For unit step input, $r(t) = u(t)$

$$R(S) = \frac{1}{S}$$

$$\therefore\ C(s) = \frac{\omega_n^2}{S^2+2\,\zeta\,\omega_n S+\omega_n^2}\left(\frac{1}{S}\right)$$

$= \frac{A}{S} + \frac{BS+D}{S^2+2\,\zeta\,\omega_n S+\omega_n{}^2}$

Here A=1, B = - 1, D = - 2 $\zeta\ \omega_n$

C(s) = $\frac{1}{S}$ - $\left(\frac{S+2\ \zeta\,\omega_n}{S^2+2\,\zeta\,\omega_n S+\omega_n{}^2}\right)$

$= \frac{1}{S} - \frac{S+\zeta\,\omega_n}{(S+2\zeta w_n\,S\,\omega_n)^2+\omega_n{}^2} - \frac{\zeta\,\omega_n}{S^2+2\,\zeta\,\omega_n S+\omega_n{}^2}$

$= \frac{1}{S} - \frac{S+\zeta\,\omega_n}{(S+\zeta\,\omega_n)^2+\omega_d{}^2} - \frac{\zeta\,\omega_n}{(S+\zeta\omega_n)^2+\omega d^2}$

Where $\omega_d = \omega_n\sqrt{1-\ \zeta^2}$

$S^2 + 2\,\zeta\,\omega_n S + \omega_n{}^2 = S^2 + 2\,\zeta\,\omega_n S + \omega_n{}^2 + \zeta^2\,\omega_n{}^2 - \zeta^2\,\omega_n{}^2$

$= (S+\zeta\omega_n)^2 + \omega_n{}^2\,(1-\ \zeta^2)$

$= (S+\zeta\omega_n)^2 + \omega_d{}^2$

Where $\omega_d = \omega_n\sqrt{1-\ \zeta^2}$

C(s) = $\frac{1}{S} - \frac{S+\zeta\,\omega_n}{(S+\zeta\omega_n)^2+\omega_d{}^2} - \frac{\zeta\,\omega_n}{(S+\zeta\omega_n)^2+\omega_d{}^2}$

Taking inverse laplace transform

C(t) = $L^{-1}(C(s))$ = 1 - $e^{-\zeta\omega_n t}$ Cos $\omega_d t$ - $\frac{e^{-\zeta\omega_n t}}{Wd}\ \zeta\ \omega_n$ sin $\omega_d t$

C(t) = 1 - $e^{-\zeta\omega_n t}$ Cos $\omega_d t$ - $\frac{e^{-\zeta Wnt}}{Wn\sqrt{1-\zeta^2}}$. $\zeta\ \omega_n$ sin $\omega_d t$.

= 1 - $\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}$ ($\sqrt{1-\ \zeta^2}$ Cos $\omega_d t$ + ζ Sin $w_d t$)

WKT, $L^{-1}\left[\frac{W}{(S+a)^2+\omega^2}\right] = e^{-at}\,sin\ \omega t$

$L^{-1}\left[\frac{S+a}{(S+a)^2+W^2}\right] = e^{-at}\,cos\ wt$

C(t) = 1 - $\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}$ Sin $(\omega_d t + \varphi)$

C(t) = 1 - $\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}$ (Sin φ Cos $\omega_d t$ + Cos φ Sin $\omega_d t$)

C(t) = 1 = $\frac{e^{-\zeta\omega nt}}{\sqrt{1-\zeta^2}}$ Sin ($\omega_d t + \varphi$)

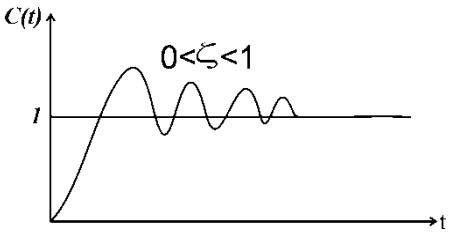
When φ = $\tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$

$c(t) = 1 \frac{e^{\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta})$

The error signal

e(t) = $c(t)$ - $r(t)$ = $\frac{e^{-\zeta\omega nt}}{\sqrt{1-\zeta^2}}$ Sin ($\omega_d t + \varphi$)

The error exhibits a damped oscillatory behavior as shown in figure.



The steady state errors , $e_{ss} = \lim_{t\to\infty} e(t) = 0$

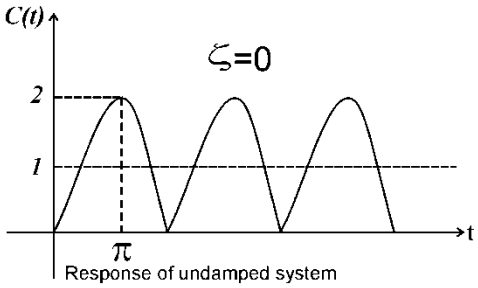
If ζ = 0, the system transient response has undamped oscillations and oscillations continues indefinitely.

If $\zeta = 0$ $\Longrightarrow$ C(t) = 1 - $\frac{e^{-0(\omega_n t)}}{\sqrt{(1-0^2}}$ Sin (ω_dt + φ)

C(t) = 1 – sin (ω_nt + 90^0)

C(t) = 1 – Cos ω_nt

The system response to unit step input when ζ = 0 shown in Fig.



Response of critically damped system to unit step input (ζ=1):

For this system, $C(s) = \frac{\omega_n^2}{S(S^2 + 2\,\omega_n S + \omega_n^2)}$

$$= \frac{A}{S} + \frac{BS+D}{S^2 + 2\,\omega_n S + \omega_n^2}$$

Where A = 1, B = - 1 and D = - 2 ω_n

$\therefore$ $C(s) = \frac{1}{S} - \frac{S+\omega_n}{S^2 + 2\,\omega_n S + \omega_n^2} - \frac{\omega_n}{S^2 + 2\,\omega_n S + \omega_n^2}$

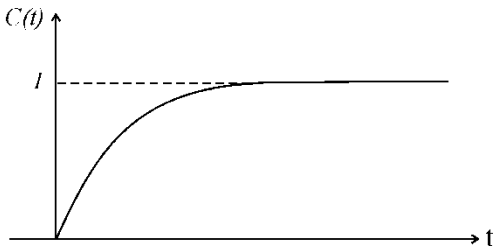
$C(s) = \frac{1}{S} - \frac{1}{S+\omega_n} - \frac{\omega_n}{(S+\omega_n)^2}$

$C(t) = 1 - Wn\, e^{-\zeta\omega_n t} . t - e^{-\zeta\omega_n t}$

$L^{-1}\,(C(s) = 1 - e^{-\omega_n t} - \omega_d t\, .\, e^{-\omega_n t}$

$\therefore$ $C(t) = 1 - e^{-\omega_n t}\,(1 + \omega_n .\, t)$

The response of the system reaches its final value without any oscillations as shown in figure.



Response of over damped system to unit step input (ζ > 1)

Here $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2}$

For unit step input $R(s) = \frac{1}{S}$

$\therefore$ $C(s) = \frac{\omega_n}{S(S^2 + 2\zeta\omega_n S + \omega_n^2)}$

$= \frac{\omega_n^2}{S(S+\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})\,(S+\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})}$

After partial fraction expansion,

$$C(s) = \frac{1}{S} + \frac{1}{2\sqrt{\zeta^2-1}\,(\zeta+\sqrt{\zeta^2-1})\,(S+\zeta\,\omega_n+\omega_n\sqrt{\zeta^2-1})}$$

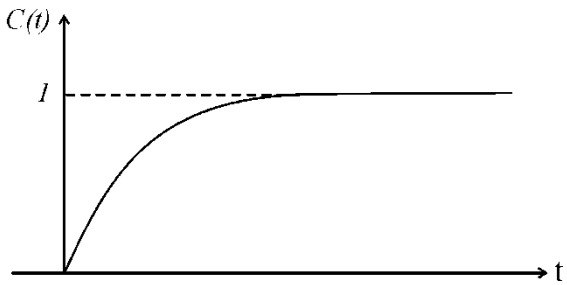
$$= \frac{1}{2\sqrt{\zeta^2-1}\,(\zeta-\sqrt{\zeta^2-1})\,(S+\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})}$$

Taking inverse LT.

$$C(t) = 1 + \frac{1}{2\sqrt{\zeta^2-1}}\left[\frac{e^{-(\zeta+\sqrt{\zeta^2-1})\,\omega_n t}}{\zeta+\sqrt{\zeta^2-1}} - \frac{e^{-(\zeta-\sqrt{\zeta^2-1}\,\omega_n t}}{(\zeta-\sqrt{\zeta^2-1})}\right]$$

From the above equation, that the transient response of the overdamped system to unit step input consists of 2 exponentially decaying curves.

The response of the system reaches its final value without any oscillations as in the case of critical damping but the response is very slow. As shown in Fig.

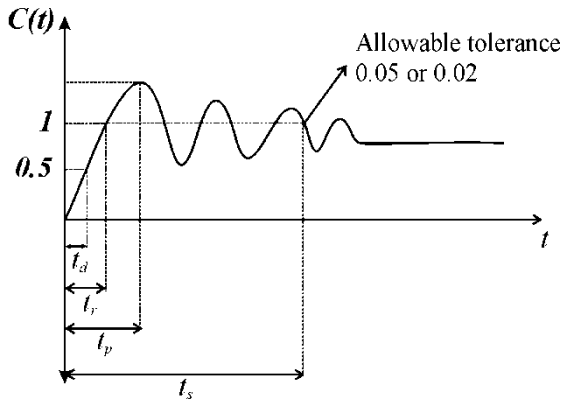


In a control system if a small amount of overshoot is acceptable, it is preferable to select valves of ζ between 0.4 and 0.8. This ensures that the system is fast and excessive overshoot is avoided.

If ζ is low, then the overshoot will be excessive. Larger values of ζ results in reduced overshoot but the system response will be slow.

4.4 Transient Specifications of second order systems :

The performance characteristics of a control system are specified in terms of the transient response to unit step input. The transient response of a practical control system exhibits damped oscillations before attaining the steady state. The response of the system is as shown in figure.



The following quantities are used to specify transient response :

1. **Delay time** , td : Defined as the time required for the response of the system to reach 50% of the final value for the very first time.

2. **Rise time** (tr): Defined as the time required for the response of the system to rise from 10% to 90%, 5% to 95% or 0% to 100% of its final value.

   For underdamaped system 0% to 100% rise time is used whereas for overdamped systems normally 10% to 90% rise time is used.

3. **Peak time tp** : The peak time 'tp' is the time required for the response to reach the first peak of the overshoot.

4. **Maximum Overshoot, Mp**: It is the maximum peak value of the response measured from unity. The overshot is an indication of the largest error between input and output during transient state. The amount of maximum overshoot indicate the relative stability of the system. If the final steady state value of the system C(∞) is different from unity.

   Maximum percent overshoot $= \frac{C(tp) - C(\infty)}{C(\infty)}$ x 100%

5. **Setting time ts** : Defined as the time required for the response to reach and stay within a specified percentage of its final value. Usually 2% or 5% criterion is used.

Derivation of Transient response specifications of 2nd order system :

I Rise time tr :

The rise time, tr is obtained when the response of the system C(t) reaches unity for the first time.

At t = tr, C(t) = 1

$$C(t) = 1 - \frac{e^{-\zeta\omega_n tr}}{\sqrt{1-\zeta^2}} \text{Sin}\left(\omega_d . \text{tr} + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right) = 1$$

Since $e^{-\zeta\omega_n tr} \neq 0$, from above equation.

$$\text{Sin}\left(\omega_d . \text{tr} + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right) = 0$$

$$\text{Wd} . \text{tr} + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta} = \pi$$

$$\text{tr} = \pi - \frac{tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_d} = \frac{\pi - \theta}{\omega_d}$$

II **<u>Peak time</u>** tp : At, t = tp, the output C(t) has a peak value $\frac{d}{dt}$ c(t) = $\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}$

$[\zeta\, \omega_n\, Sin\,(\omega_d t + \theta) - \omega_d\, Cos\,(\omega_d t + \theta)]$

At t = tp, $\frac{d\,c(t)}{dt} = 0$

$$\frac{dc\,(tp)}{dt} = \frac{e^{-\zeta\omega_n tp}}{\sqrt{1-\zeta^2}}\,[\zeta\omega_n\, Sin\,(\omega_d tp + \theta) - \omega_d\, Cos\,(\omega_d tp + \theta)] = 0$$

Since $e^{-\zeta\omega_n tp} \neq 0$

$\zeta\, \omega_n$ Sin ($\omega_d$tp + θ) - ω_d Cos ($\omega_d$tp + θ) = 0

$\zeta\, \omega_n$ Sin ($\omega_d$tp + θ) - $\omega_n \sqrt{1-\zeta^2}$ Cos ($\omega_d$tp + θ) = 0

Sin ($\omega_d$tp + θ – θ) = 0

Sin ($\omega_d$tp) = 0

ω_d . tp = π Or $tp = \frac{\pi}{\omega_d}$

Note : In general the condition ω_d.t = n π gives the time at which $\frac{d}{dt}$ c(t) = 0.

III Maximum Overshoot Mp :

The maximum overshoot occurs at the peak time,

$$tp = \frac{\pi}{\omega_d}$$

Mp = C(tp) – 1

$$= -\frac{e^{-\zeta\omega_n tp}}{\sqrt{1-\zeta^2}} \text{ Sin } (\omega_d \text{tp} + \theta)$$

But $\omega_d$tp = π

$$Mp = -\frac{e^{-\zeta\omega_n tp}}{\sqrt{1-\zeta^2}} \text{ Sin } (\pi - \theta)$$

$$= \frac{e^{-\zeta\omega_n tp}}{\sqrt{1-\zeta^2}} \text{ Sin } \theta$$

But Sin $\theta = \sqrt{1-\zeta^2}$ $\quad \therefore$ Mp $= e^{-\frac{\omega_n \pi}{\omega_d}} = e^{-\frac{\omega_n \pi}{\omega_n\sqrt{1-\zeta^2}}} = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$

Maximum % Overshoot $= e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$ x 100%

IV **Setting time ts** : ts gives an indication of the time required for the system response to reach and stay within a specified percentage of its steady state value (usually 2% or 5%).

The output of underdamped system to a unit step input is

$$c(t) = 1 \pm \frac{e^{-\zeta\omega_n t.}}{\sqrt{1-\zeta^2}} \text{ Sin } (\omega_d \text{t} + \theta)$$

It can be observed from the above equation, that the output is always bounded by two exponentially decaying curves

$$\left(1 + \frac{e^{-\zeta\omega_n t.}}{\sqrt{1-\zeta^2}}\right) \ \& \ \left(1 - \frac{e^{-\zeta\omega_n t.}}{\sqrt{1-\zeta^2}}\right)$$

Time constant of these curve is T = $\frac{1}{\zeta\omega_n}$

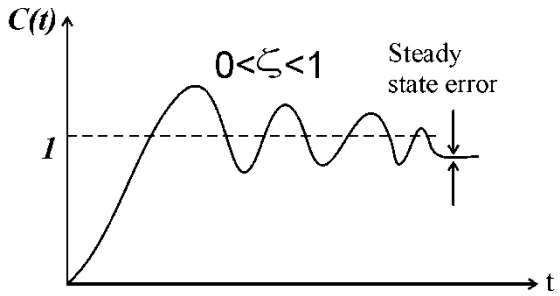
$$\omega_n ts = \frac{4}{\zeta} \therefore \text{ ts} = \frac{4}{\zeta\omega_n} = 4\text{T}.$$

For 2% criterion, setting time ts = 4T

Steady State Response :

When a certain input is given to a control system, the system response usually has transient portion. The response of the system when the transients have become negligible is known as the steady state response. It is desirable to have steady state response equal to the reference input signal. But in practice due to system imperfections, the steady state output may not match with the reference input. In such a case the system is said to have a steady state error.

Fig. shows the steady state error of a system subjected to step input.



The upper envelop curve reaches to 5% of the final value of the output at t = ts if 5% criterion used. The value of the upper envelop at this instant is 1.05.

$$1 + \frac{e^{-\zeta\omega_n ts}}{\sqrt{1-\zeta^2}} = 1.05$$

$$\frac{e^{-\zeta\omega_n .ts}}{\sqrt{1-\zeta^2}} = 0.05$$

From the above equation ω_n.ts = $-\frac{1}{\zeta}$ log e $(0.05\sqrt{1-\zeta^2})$

For small value of ζ in the above equation is approximated to ω_n.ts = $\frac{3}{\zeta}$

$$\therefore \text{ ts} = \frac{3}{\zeta.\omega_n} = 3T$$

Same result is obtained if the lower envelop is used. For 5% criterion, settling time ts = 3T. For 2% criterion, the upper envelop has a value of 1.02 at t = ts.

$$1 + \frac{e^{-\zeta\omega_n ts}}{\sqrt{1-\zeta^2}} = 1.02$$

From the above equation $e^{-\zeta\omega_n ts} = 0.02\sqrt{1-\zeta^2}$

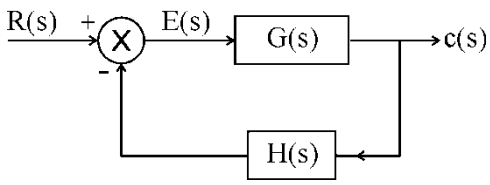
In figure, it is seen that steady state response is not equal to the reference input.

In case of unity feedback system H(s) = 1, the error signal is e(t) = r(t) – c(t). In case of non unity feed back system, the error signal is e(t) = r(t) - b(t),

Where b(t) is the feedback signal.

Type of a Control systems :

A simple closed loop feedback system in figure. The open loop transfer function G(s) H(s) is dependent on the components of the control system.



Depending on the system the open loop transfer function has one of the following forms.

$$G(s)\,H(s) = \frac{K(T_aS+1)\,(T_bS+1)}{S^0(T1S+1)\,(T2s+1)} \quad \ldots\ldots \quad (1)$$

$$G(s)\,H(s) = \frac{K(T_aS+1)\,(T_bS+1)}{S(T1S+1)\,(T2s+1)} \quad \ldots\ldots \quad (2)$$

$$G(s)\,H(s) = \frac{K(T_aS+1)\,(T_bS+1)}{S^2(T1S+1)\,(T2s+1)} \quad \ldots\ldots \quad (3)$$

In general,

$$G(s)\,H(s) = \frac{K(TaS+1)\,(Tbs+1)}{S^N(T1S+1)\,(T2s+1)} \quad \ldots\ldots \quad (4)$$

Here K is the gain and N is known as the type of the system. The terms S^N represents the number of poles of G(s) H(s) at the origin.

A system is called type 0, type 1, type 2, if N = 0, 1, 2 respectively. The system of equation (1) is a type 0 system, the system of equation (2) is a type 1 system and the system of equation (3) is a type 2 system.

It is interesting to note that steady state error is not dependent on the order of the system. But is determined by the number of poles at origin of the open loop transfer function.

It is found that increase in the type results in reduction of steady state error but stability of the system is affected.

4.5 Steady state errors :

The closed loop transfer function is shown in above figure.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

Here, the error signal E(s) = R(s) – B(s)

E(s) = R(s) – H(s) C(s)

C(s) – G(S) E(S)

∴ E(s) = R(s) – H(s) G(s) E(s)

E(S) (1G(s) H(s)) = R(s)

E(s) (1+G(s) H(s)) = R(s)

$$E(s) = \frac{R(s)}{1+G(s)H(s)}$$

Substituting for C(s) in the above equation.

$$E(s) = R(s) - \frac{H(s)\,G(s)\,R(s)}{1+G(s)\,H(s)}$$

$$= \frac{R(s)\,[1+G(s)H(s)-G(s)H(s)]}{1+G(s)\,H(s)}$$

$$E(s) = \frac{1}{1+G(s)\,H(s)}\,R(s)$$

By using final value theorem, the steady state error of the system e_{ss} is determined.

Types of Steady State error :

1) Static error constants (Given error for the standard inputs).

2) Dynamic error constants (Given error for any inputs).

Steady State error due to step input :

When a system is subjected to step input r(t) = A . u(t)

$$ess = \lim_{s\to 0} \frac{S\,R(s)}{1+G(s)\,.H(s)}$$

$$R(s) = \frac{A}{S}$$

$\therefore$ $e_{ss} = \lim_{s\to 0} \frac{S.\ (A/s)}{1+G(s)\ .H(s)} = \frac{A}{1+\lim_{s\to 0} G(s)\ .H(s)}$

$e_{ss} = \frac{A}{1+Kp.}$ where Kp = Static position error constant

From equation it can be concluded that a system subjected to step input has zero steady state error if Kp is ∞. Otherwise, system has steady state error.

For type 0 system, G(s) H(s) $= \frac{K\ (T_a s+1)\ (T_b s+1)}{(T_1 s+1)\ (T_2 s+1)}$

$\therefore$ Kp $= \lim_{s\to 0} G(s).H(s) = \lim_{s\to 0} \frac{K\ (T_a s+1)\ (T_b s+1)}{(T_1 s+1)\ (T_2 s+1)}$

Kp = K

For a type 1 or higher system,

Kp $= \lim_{s\to 0} \frac{K\ (Tas+1)\ (Tbs+1)}{(T1S+1)\ (T2S+1)} = \infty$ (N $\geq$ 1)

Hence

For type 0 system, $e_{ss} = \frac{A}{1+Kp} = \frac{A}{1+K}$

For type 1 and higher systems, $e_{ss} = \frac{A}{1+Kp} = 0$

Static portion error constant Kp :

Kp $= \lim_{s\to 0} G(s)\ H(s)$ for unit step input

G(s) H(s) $= \frac{K\ (1+ST1)\ (1+ST2)\ldots}{S^N\ (1+STa)\ (1+STb)\ldots}$

for N = 0, type 0 system

N = 1 type 1 system

But E(s) $= \left\{\frac{1}{1+G(s)\ H(s)}\right\}$ R(s)

But R(s) $= \frac{1}{s}$ unit step input

$\therefore$ E(s) $= \left\{\frac{1}{1+G(s)\,H(s)}\right\}\frac{1}{S}$

Steady state error

$e_{ss} = \lim_{S\to 0} S\,.E(s)$

$e_{ss} = \lim_{S\to 0} S\ \left\{\frac{1}{1+G(s)\,H(s)}\right\}\frac{1}{S}$

$= \frac{1}{1+\lim_{S\to 0} G(s)\,H(s)}$

$\therefore$ $e_{ss} = \frac{1}{1+Kp}$

Type 0 system :

G(s) H(s) $= \frac{K\,(1+ST1)\,(1+ST2)\ldots}{S^0\,(1+STa)\ldots}$

Kp $= \lim_{S\to 0} G(s)H(s) = K$

$e_{ss} = \frac{1}{1+Kp} = \frac{1}{1+K}$ For type 0 system

Type 1 system :

G(s) H(s) $= \frac{K(1+ST1)\ldots.}{S^1\,(1+ST2)\ldots.}$

Kp $= \lim_{S\to 0} G(s)H(s) = \infty$

$\therefore$ $e_{ss} = \frac{1}{1+Kp} = \frac{1}{1+\infty} = 0$

Type 1 and above, $e_{ss} = 0$

Steady State error due to Ramp input (State Velocity error constant K_v):

Ramp input r(t) = A.t. u(t), R(s) $= \frac{A}{S^2}$

$e_{ss} = \lim_{S\to 0}\frac{S.R(s)}{1+G(s)\,H(s)} = \lim_{S\to 0}\frac{S\left(A/_{S^2}\right)}{1+G(s)\,H(s)}$

$$= \lim_{s\to 0} \frac{A}{S+S.G(s)H(s)} = \frac{A}{0+\lim_{s\to 0} S.G(s)\,H(s)}$$

$$e_{ss} = \frac{A}{Kv}$$

Where Kv is known as Velocity error constant and is given by

$$Kv = \lim_{s\to 0} S.G(s)H(s)$$

For a type 0 system,

$$Kv = \lim_{s\to 0} S\,G(s)H(s)$$

$$= \lim_{s\to 0} \frac{S\,K\,(1+ST1)\ldots..}{S^0\,(1+STa)\ldots..}$$

$$Kv = 0$$

For type 1 system :

$$Kv = \lim_{s\to 0} \frac{SK\,(Tas+1)\,(Tbs+1)\ldots.}{S(T1S+1)\,(T2S+1)\ldots} = K$$

$$Kv = K$$

For type 2 system and higher systems,

$$Kv = \lim_{s\to 0} \frac{S\,K\,(Tas+1)\,(Tbs+1)\ldots.}{S^N(T1S+1)\,(T2S+1)\ldots} = \infty \quad \underline{N \geq 2}$$

For type 0 system, $e_{ss} = A/Kv = \infty$

For type 1 system $e_{ss} = A/Kv = A/K$

For type 2 and higher systems $e_{ss} = A/Kv = 0$

Steady state error due to parabolic input (Ka)

Here $r(t) = \frac{A}{2}\, t^2\, u(t)$

$R(s) = A/2\ .\ 2/S^3 = A/S^3$

$$e_{ss} = \lim_{s\to 0} \frac{s.R(s)}{1+G(s)\,H(s)} = \lim_{s\to 0} \frac{S\,(A/S^3)}{1+G(s)\,H(s)}$$

$$= \lim_{s\to 0} \frac{A}{S^2 + S^2\, G(s)\, H(s)} = \frac{A}{0 + \lim_{S\to 0} s^2\, G(s)\, H(s)} = \frac{A}{Ka}$$

Where Ka $= \lim_{s\to 0} S^2\, G(s) H(s)$ and is known as acceleration error const.

For type 0 system, $K_a = \lim_{s \to 0} s^2\, G(s) H(s)$

$$\text{Ka} = \lim_{s\to 0} \frac{S^2 K\,(Tas+1)\,(Tbs+1)}{(T1S+1)\,(T2S+1)} = 0$$

For type 1 system,

$$\text{Ka} = \lim_{s\to 0} S^2\, G(s)\,.\,H(s) = \lim_{s\to 0} \frac{S^2 K\,(Tas+1)\,(Tbs+1)}{S(T1S+1)\,(T2S+1)} = 0$$

For type 2 system,

$$\text{Kv} = \lim_{s\to 0} S^2\, G(s)\, H(s)$$

$$= \lim_{s\to 0} \frac{S^2 K\,(Tas+1)\,(Tbs+1)}{S^2(T1S+1)\,(T2S+1)} \;\ldots..$$

<u>Kv = K</u>

Imported to remember

II order system :

$$\text{T(s)} = \frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S^2 + 2\,\zeta\,\omega_n S + \omega_n{}^2}$$

Underdamped

$$\text{C(t)} = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\,(\omega_d \text{t} + \theta) \quad 0 < \zeta < 1$$

$$\theta = \tan^{-1} \frac{1-\zeta^2}{\zeta},\; \omega_d = \omega_n \sqrt{1-\zeta^2}$$

Undamped

$$\text{C(t)} = 1 - \text{Cos}\, \omega_n \text{t}.$$

$$\text{rise time (tr)} = \frac{\pi - \theta}{\omega_d}$$

$$\text{Peak time (tp)} = \frac{\pi}{\omega_d}$$

Maximum overshoot Mp :

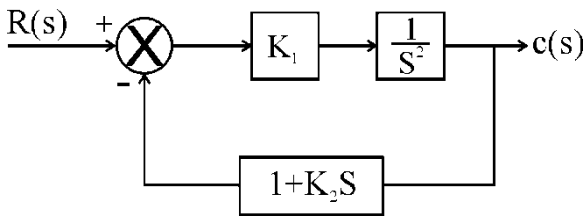
$$\%\ \text{Mp} = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

Settling time ts $= \frac{4}{\zeta\omega_n}$

Delay time (Td) $= \frac{1+0.7\,\zeta}{\omega_n}$

Ex. Consider the servo mechanism shown in figure. Determined values of K1 and K2 so that the maximum overshoot in unit step response is 20% and peak is 3 Sec.

Solution :



Here $\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)\,H(s)}$

G(s) $= \frac{K1}{S^2}$. H(s) $= 1 + k_2 s$

$$\frac{C(s)}{R(s)} = \frac{K1/S^2}{1+\left(\frac{K1}{S}^2\right)(1+K2S)} = \frac{K1}{S^2+K1+K1K2S}$$

$$\frac{C(s)}{R(s)} = \frac{K1}{S^2+K1K2S+K1}$$

The equation is in the form of

$$\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S^2+2\,\zeta\omega_n S+\omega_n{}^2}$$

Where K1 = $\omega_n{}^2$, K1K2 = 2 $\zeta\omega_n$

Here Mp = 0.2, tp = 3 Sec Mp = $e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$ = 0.2

Taking natural logarithm on both sides

$-\frac{\pi\zeta}{\sqrt{1-\zeta^2}} = -1.609 \quad \therefore\ \zeta = 0.456$

Peak time tp $= \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

$$\omega_n = \frac{\pi}{tp\sqrt{1-\zeta^2}} = \frac{\pi}{3\ \sqrt{1-(0.456)^2}} = 1.176$$

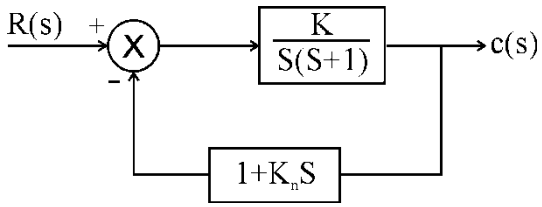
$\therefore\ \omega_n = 1.176$ rad/sec and $\zeta = 0.456$

Substituting K1 = $\omega_n{}^2 = (1.176)^2 = 1.383$

And K2 $= \frac{2\zeta\omega_n}{K1} = \frac{2 \text{ x } 0.456 \text{ x } 1.176}{1.383}$

<u>K2 = 0.7738</u>

Ex. For the system given in figure below, determine the values of K and Kn so that Mp is 20% and peak time tp = 1 sec. Also obtain rise time tr and settling time ts for these values.



Solution:

$$\frac{C(s)}{R(s)} = \frac{\frac{K}{S(S+1)}}{1+\frac{K}{S\,(S+1)}\,1+KnS} = \frac{K}{S^2+\,(K.Kn+1)S+K}$$

Comparing with the transfer function of a II order system,

$$\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S^2+\,2\zeta\omega_n S+\omega_n{}^2}$$

We get, K = $\omega_n{}^2$, 2 $\zeta\omega^2 n$ $\zeta\omega_n$ = K. Kn + 1

To determine ζ

$$\text{Mp} = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$0.2 = \dot{e}\,\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}$$

Taking natural logarithm on both sides of

$$\frac{-\pi\zeta}{\sqrt{1-\zeta^2}} = -\,1.6 \quad \zeta = 0.456$$

$$\text{tp} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n.\sqrt{1-\zeta^2}} \qquad \therefore\ \omega_n = \frac{\pi}{tp\sqrt{1-\zeta^2}}$$

For tp = 1 sec

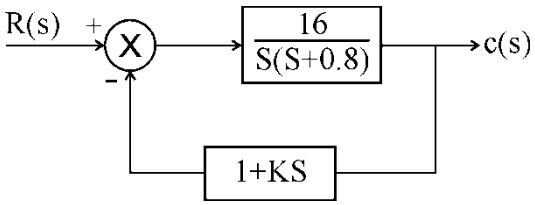
$$\omega_n = \frac{\pi}{\sqrt{1-\zeta^2}} = 3.53 \text{ rad/sec}$$

K = $\omega_n{}^2$ = 12.46 Kn = $\frac{2\zeta\omega_n - 1}{K}$ = $\frac{2(0.456)(3.53) - 1}{12.46}$

= <u>K = 12.46</u> <u>Kn = 0.1781</u>

Ex. For the system shown in figure below, determine the value of K, so that damping ratio is 0.5. Also obtain rise time tp, maximum overshoot and settling time for unit step input.

Solution :



$$\frac{C(s)}{R(s)} = \frac{16}{S^2 + (16K+0.8)S + 16}$$

Comparing

$$\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S^2 + 2\zeta\,\omega_n S + \omega_n{}^2}, \qquad \omega_n{}^2 = 16,$$

$$\omega_n = 4 \text{ rad/sec}$$

$2\,\zeta\,\omega_n$ = 16K + 0.8 or

K = $\frac{2\,\zeta\,\omega_n - 0.8}{16}$ and ζ = 0.5 (given)

K = $\frac{2\,(0.5)4 - 0.8}{16}$ = 0.2

Rise time tr = $\frac{n - \theta}{\omega_n\sqrt{1-\zeta^2}}$, $\theta = \frac{tan^{-1}\sqrt{1-\zeta^2}}{\zeta}$ = 1.047 rad

∴ tr = $\frac{\pi - 1.047}{4\sqrt{1-(0.5)^2}}$ = 0.604 sec.

tp = $\frac{\pi d}{\omega_d} = \frac{\pi}{3.464}$ = 0.906 sec.

Max. Overshoot = Mp = $e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = e^{\frac{-\pi\,(0.5)}{\sqrt{1-(0.5)^2}}}$ = 0.163

For 2% criterion For 5% criterion

ts = $\frac{4}{\zeta\,\omega_n}$ = 2 sec. ts = $\frac{3}{\zeta\,\omega_n}$ = 1.5 sec.

Ex. Negative feed back system has a forward path transfer function = $\frac{K}{S(S+1)}$ while the feedback path transfer function is (1 + as). If this system is to have a peak time of 0.5 sec and also a 10% overshoot, for a unit step input determine K & a.

Solution :

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)\,H(s)}$$

$$= \frac{\frac{K}{S(S+1)}}{1+\frac{K}{S\,(S+1)}(1+as)}$$

$$= \frac{K}{S^2+(Ka+1)S+K} \quad \ldots\ldots\ldots (1)$$

Comparing equation (1) with the transfer function of a II order system.

$$\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S^2+2\zeta\omega_n S+\omega_n{}^2}$$

$$K = \omega_n{}^2,\ (Ka + 1) = 2\zeta\omega_n \qquad - \qquad (2)$$

To determine ζ

$$Mp = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \Longrightarrow 0.1 = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \qquad - \qquad (3)$$

Taking Natural logarithm on both sides of equation (3)

$$\frac{-\pi\zeta}{\sqrt{1-\zeta^2}} = -2.303$$

$$\pi^2\zeta^2 = 5.3\,(1-\zeta^2)$$

$$\zeta = \mathbf{0.592}$$

To determine ω_n

$$tp = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$0.5 = \frac{\pi}{\omega_n\sqrt{1-(0.592)^2}}$$

$\omega_n = \frac{\pi}{0.5\sqrt{0.651}} = 7.786$ rad/sec

K = $\omega_n^2 = 60.62$

Ka + 1 = 2 ζ ω_n

Ka + 1 = 2 (0.592) (7.786) = 9.22

a = $\frac{9.22-1}{K} = \frac{8.22}{60.62} = 0.136$

a = 0.136

Ex. The closed loop transfer function of a 2nd order system is $\frac{C(s)}{R(s)} = \frac{25}{S^2+65+25}$. Find rise time, peak time, maximum overshoot and settling time, if the system is subjected to unit step input. Assume allowable steady state error as 2%. Find also its output response.

Solution:

$\frac{C(s)}{R(s)} = \frac{25}{S^2+65+25}$

$\frac{C(s)}{R(s)} = \frac{\omega_n}{S^2+2\zeta\omega_n S+\omega_n^2}$ ⟸ General form of II order system.

Comparing both $\omega_n^2 = 25$, $\omega_n = 5$ rad/sec

$2\zeta\omega_n = 6$, $\zeta = \frac{6}{2\,\omega_n} = \frac{6}{2 \times 5} = \frac{6}{10} = 0.6$

$\theta = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$ $\theta = 0.927$ rad.

tr = $\frac{\pi-\theta}{\omega_n\sqrt{1-\zeta^2}}$

$= \frac{\pi-0.927}{5\sqrt{1-0.6^2}} = 0.5354$ sec.

tp = $\frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \pi/4 = 0.7854$ sec.

Mp = $e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.0948$ % Mp = 9.48

For 2% criterion

ts $= \frac{\pi}{\zeta\omega_n} = \frac{4}{0.6 \text{ x } 5} = 1.333$ sec.

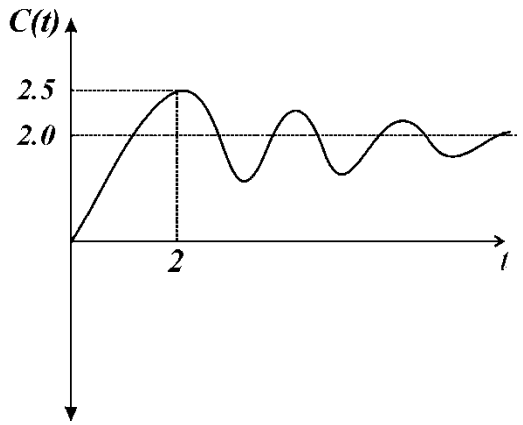
c(t) $= 1 - \frac{e^{\zeta\omega_n t}}{1-\zeta^2} \sin\ (w_d t + 0)$

c(t) = 1 – 1.25 e^{-3t} Sin (4t + 0.927)

M $= \frac{K}{\omega_n{}^2} = \frac{500}{(1.3)^2} = 291.35$ kg.

B = (2 $\zeta\omega_n$) (M) = 458 N/M/Sec.

Ex. The step response of a II order system is shown in figure for an input of 2*u*(*t*). Determine the open loop and closed loop transfer functions. Assume unity feedback system.



Solution

From Figure tp = 2 sec.

Mp = 0.25 at steady state

% Mp $= \frac{C(tp) - C(\infty)}{C(\infty)} = 0$

$\pi\zeta = 1.386\sqrt{1-\zeta^2}$

Squaring on both side

$\pi^2\,\zeta^2 = 1.921\,(1 - \zeta^2)$

$\zeta^2\,.\,(11.791) = 1.921$

$\zeta^2 = \frac{1.921}{11.79} = 0.163$ $\qquad$ $\boldsymbol{\zeta = 0.4036}$

tp $= \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$

$$2 = \frac{\pi}{\omega_n\sqrt{1-(0.4036)^2}} = 2(\omega_n\sqrt{1-(0.4036)^2} = \pi$$

$$\omega_n{}^2\,(1-0.163)\,4 = \pi^2$$

$$\omega_n{}^2 = \frac{\pi^2}{3.348} = 2.948$$

$$\therefore\ \omega_n = 1.717 \text{ rad/sec}$$

Ex. - The open loop transfer function of a unity feedback control system is given by

$G(s) = \frac{K}{S(ST+1)}$, find

(i) By what factor should K is multiplied so that damping ratio is reduced from 0.2 to 0.8.

(ii) By what factor should T be multiplied so that damping ratio is reduced from 0.6 to 0.3.

Solution:

$$G(s) = \frac{K}{S(ST+1)} \quad H = 1$$

$$\frac{C}{R} = \frac{G}{1+GH} = \frac{\frac{K}{S(ST+1)}}{1+\underbrace{\frac{K}{S(ST+1)}-1}_{S^2T+S}} = \frac{K}{S^2T+S+K}$$

$$\frac{K}{S^2T+S+K}$$

$$\frac{C}{R} = \frac{\frac{K}{S(ST+1)}}{\frac{S^2T+S+K}{S(ST+1)}} = \frac{K}{S^2T+S+K} \div \text{ by } T.$$

$$\frac{C}{R} = \frac{K/T}{S^2+S/T+K/T}$$

Comparing dominator $(S^2 + S/T + K/T)$ with $(S^2 + 2\zeta\omega_n S + \omega_n{}^2)$

$$\therefore\ \omega_n{}^2 = K/T\,,\ 2\omega_n\,\zeta = \frac{1}{T}$$

Here $\zeta = \frac{1}{2\omega_n T}$ But $\omega_n = \sqrt{\frac{K}{T}}$

$$\zeta = \frac{1}{2\sqrt{T}\,.\frac{\sqrt{K}}{\sqrt{T}}\,.\,\sqrt{T}} = \frac{1}{2\sqrt{KT}}$$

Here $\zeta 1 = 0.2 = \frac{1}{2\sqrt{K1T1}} \Longrightarrow 0.4 = \frac{1}{\sqrt{K1T1}}$

Now $\zeta 2 = 0.8 = \frac{1}{2\sqrt{K2T2}} \Rightarrow 1.6 = \frac{1}{\sqrt{K2T2}}$

But T2 = T1 squaring

$$\frac{\frac{1}{\sqrt{K1T1}}}{\frac{1}{\sqrt{K2T2}}} = \frac{0.4}{1.6} \Rightarrow \frac{\sqrt{K2T2}}{\sqrt{K1T1}} = 0.25 \Rightarrow \frac{K2\,.T2}{K1\,.T1} = 0.0625$$

$$\frac{K2}{K1} = 0.0625$$

Thus K2 = 0.0625, K1 = 1

Thus reduce gain 0.0625 times previous value.

(ii) $\zeta = 0.6 = \frac{1}{2\sqrt{K1T1}} \Rightarrow 0.6 \text{ x } 2 = \frac{1}{\sqrt{K1T1}}$

$\zeta 2 = 0.3 = \frac{1}{2\sqrt{K2T2}} \Rightarrow 0.3 \text{ x } 2 = \frac{1}{\sqrt{K2\,.T2}} \cdot \frac{T2}{T1} = 1$

But now K1 = K2

$$\therefore \frac{\zeta\,1}{\zeta 2} = \frac{0.6}{0.3} = \sqrt{\frac{T2}{T1}}$$

Ex. Measurements conducted in a servo mechanism show the system response to be C(t) = 1 + 0.2 $e^{-60t} - 1.2\, e^{-10t}$ when subjected to a unit step input. Obtain the expression for closed loop transfer function the damping ratio and undamped natural frequency of oscillation.

$$c(t) = 1 + 0.2\, e^{-60t} - 1.2\,.\, e^{-10t}$$

Taking L.T. of *c*(*t*).

$$\frac{C(s)}{R(s)} = \frac{1/S + \frac{0.2}{S+60} - \frac{1.2}{S+10}}{1/S} = \frac{1+0.25}{S+60} - \frac{1,.2S}{S+10}$$

$$= \frac{(S+10)(S+60) + 0.2S\,(S+10) - 1.25\,(S+60)}{(S+60)(S+10)}$$

$$= \frac{600}{S^2 + 70S + 600} \rightarrow \text{Closed loop Transfer function.}$$

$\therefore\ \omega_n^2 = 600$

$\omega_n = 24.495$ rad/sec and $2\zeta\omega_n = 70$

$\zeta = \frac{70}{2.\omega_n} = 1.4288$

Ex. Identify the type of the System

(i) $G(s) = \frac{K(1+3S)}{S^2}$ $H(s) = \frac{1+5S}{S^2+6S+7}$

$G(s)\,H(s) = \frac{K(1+3S)(1+5S)}{S^2\,(S^2+6S+7)}$

This is type 2 System.

(ii) $G(s) = \frac{K}{S}$ and $H(s) = \frac{1}{S\,(S+2)}$

$G(s)\,H(s) = \frac{K\ (1)}{S\ S(S+2)} = \frac{K}{S^2\,(S+2)}$

This is a type 2 s/m.

(iii) $G(s) = \frac{4}{S^2+6S+7}$ and $H(s) = (S+3)$

$G(s)\,H(s) = \frac{4\,(S+3)}{S^2+6S+7}$

This is type zero system.

(iv) $G(s) = \frac{5}{S}$ and $H(s) = \frac{1}{S}$

$G(s)\,H(s) = \frac{5}{S^2}$ This is a type 2 system

Ex. Find the static error coefficients for a system having

$G(s)\,H(s) = \frac{(S+3)}{S\,(1+0.60\,S)(1+0.35\,S)}$

Solution:

$$Kp = \lim_{S\to 0} G(s)H(s) = \lim_{S\to 0} \frac{(S+3)}{S\,(1+0.60\,S)(1+0.35\,S)}$$

$Kp = \infty$

$$Kv = \lim_{S\to 0} S\,G(s)H(s) = \lim_{S\to 0} \frac{\cancel{S}\,(S+3)}{\cancel{S}\,(1+0.60\,S)(1+0.35\,S)}$$

$Kv = 3$

$$Ka = \lim_{s\to 0} S^2 G(s)H(s) = \lim_{s\to 0} \frac{S\,(S+3)}{(1+0.60\,S)(1+0.35\,S)}$$

$$Ka = 0$$

Ex. Find Kp, Kv, Ka for the system having $G(s) = \frac{10}{S^2}$ and $H(s) = 0.7$

Solution:

$$Kp = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0}\left(\frac{10}{S^2}\right)(0.7) = \infty$$

$$Kv = \lim_{s\to 0} S\,G(s)\,H(s)$$

$$= \lim_{s\to 0}\left\{S\left(\frac{10}{S^2}\right)0.7\right\}$$

$$Kv = \infty$$

$$Ka = \lim_{s\to 0} S^2\,G(s)H(s)$$

$$= \lim_{s\to 0}\left\{S\left(\frac{10}{S^2}\right)0.7\right\}$$

$$Ka = 7$$

Ex. Find Kp, Kv and Ka for the system having $G(s) = \frac{5}{S^2+3S+5}$ and $H(s) = 0.6$

Solution:

$$Kp = \lim_{s\to 0} G(s)\,H(s)$$

$$= \lim_{s\to 0}\left(\frac{5}{S^2+3S+5}\right)(0.6)$$

$$Kp = 0.6$$

$$Kv = \lim_{s\to 0} S\,G(s)\,H(s) = \lim_{s\to 0}\left(\frac{3s}{S^2+3S+5}\right) = 0$$

$$Ka = \lim_{s\to 0} S^2\,G(s)\,H(s) = \lim_{s\to 0}\left(\frac{3S^2}{S^2+3S+5}\right) = 0$$

Ex. For a S/m having $G(s)\,H(s) = \frac{K\,(S+4)}{S\,(S^3+5\,S^2+6S)}$ find

(i) type of the system (ii) error coefficient, (iii) error due to input $\frac{A}{2}t^2$

Solution:

$$G(s)\,H(s) = \frac{K\,(S+4)}{S\,(S^3+5\,S^2+6S)} = \frac{K\,(S+4)}{S^2\,(S^2+5\,S+6)}$$

(i) This is a type 2 system

Since it is type 2 system

Kp and Kv = ∞

(ii) Ka = $\lim_{S\to 0} S^2\, G(s)\, H(s)$

$= \lim_{S\to 0} \left\{ \cancel{S^2} \frac{K\,(S+4)}{\cancel{S^2}\,(S^2+5\,S+6)} \right\}$

$= \frac{4\,K}{6} = \frac{2\,K}{3}$

(iii) $e_{ss} = \frac{A}{K_a} = \frac{A}{\frac{2K}{3}} = \frac{3A}{2K}$

Ex. A unity feedback system has G(s) = $\frac{180}{S\,(S+6)}$ and r(t) = 4 t

Determine (i) the steady state error

(ii) the value of K to reduce the error by 6%

Solution :

G(s) H(s) = $\frac{180}{S\,(S+6)}$

r(t) = 4 t

R(s) = $\frac{4}{S^2}$

$e_{ss} = \lim_{S\to 0} \frac{S\,R\,(s)}{1+G(s)\,H(s)}$

$= \lim_{S\to 0} \frac{S\,.\,\frac{4}{S^2}}{1+\frac{180}{S\,(S+6)}} = \lim_{S\to 0} \frac{4}{S+\frac{180}{S+6}}$

e_{ss} = 0.134

Since new error is 6% less,

$(e_{ss})_{\text{new}}$ = (0.94) e_{ss} = 0.12596

$e_{ss} = \lim_{S\to 0} \left[\frac{4}{S+\frac{K}{S+6}}\right] = \frac{24}{K}$

$\frac{24}{K}$ = 0.12596

∴ K = $\frac{24}{0.12596}$ = 190.537

Static error constants

Constant	Equation	Steady State error (e_{ss})
Position error constant (Kp)	Kp = $\lim_{s\to 0}$ S G(s) H(s)	$e_{ss} = \frac{A}{1+Kp}$
Velocity error constant (Kv)	Kv = $\lim_{s\to 0}$ S G(s) H(s)	$e_{ss} = \frac{A}{Kv}$
Acceleration error constant (Ka)	Kv = $\lim_{s\to 0} S^2$ G(s) H(s)	$e_{ss} = \frac{A}{Ka}$

<u>Error for different types of input</u>

Type	Step input		Ramp input		Parabolic input	
	Kp	e_{ss}	Kv	e_{ss}	Ka	e_{ss}
Type 0	K	$\frac{A}{1+K}$	0	∞	0	∞
Type 1	∞	0	K	$\frac{A}{K}$	0	∞
Type 2	∞	0	∞	0	K	$\frac{A}{K}$

Sl.No.	Time response specification
1.	Delay time $t_d = \frac{1+0.7\,\zeta}{\omega_n}$
2.	$t_r = \frac{\pi-\theta}{\omega_d} \Longrightarrow$ rise time Where $\theta = \frac{tan^{-1}\sqrt{1-\zeta^2}}{\zeta}$ in radian. $\omega_d = \omega_n\sqrt{1-\zeta^2}$
3.	Peak time tp $= \frac{\pi}{\omega_d}$
4.	Maximum Overshoot % Mp $= e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$ x 100
5.	Settling time

	ts $= e^{-4/\zeta\omega_n}$ for 2% criterion Or ts $= e^{-2/\zeta\omega_n}$ for 5% criterion

Ex. For a S/m having $\frac{C(s)}{R(s)} = \frac{20}{S^2+7S+2S}$ find its time response specifications and expression for output.

Solution:

$\frac{C(s)}{R(s)} = \frac{2\,o}{2\,S}\left[\frac{2\,S}{S^2+7\,S+2\,S}\right]$ comparing with standard for $\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{S^2+2\zeta\omega_n S+\omega_n^2}$

$\omega_n^2 = 25$, $\omega_n = 5$ rad/sec

$2\,\zeta\,\omega_n = 7$

$\zeta = \frac{7}{10} = 0.7$

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 5\sqrt{1-(0.7)^2} = 3.57$ rad/sec.

td $= \frac{1+0.7\,\zeta}{\omega_n} = 0.298$ sec.

tr $= \frac{\pi-\theta}{\omega_d}$

$\theta = \tan^{-1}\frac{\zeta}{\sqrt{1-\zeta^2}} = 0.775$ rad.

tr $= 0.662$ sec.

tp $= \frac{\pi}{\omega_d} = 0.879$ sec.

Mp $= 100\,xe^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = 4.60\%$

$t_S = \frac{4}{\zeta\omega_n} = 1.1428$ sec.

For unit step input

$$y(t) = \frac{20}{25}\left(1-\frac{e^{-3.5\,t}}{0.714}\,Sin\,(3.57t+0.775\right)$$

Ex. Determine the time response specifications and expression for output for unit step input to the system having system equations as follows.

Solution:

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 16\,y = 9\,x$$

Where y is output and x is input

Solution : $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 16\,y = 9\,x$

Taking LT on both sides.

$S^2\,Y(s) + 5\,S\,Y(s) + 16\,Y(s) = 9\,X(s)$

$(S^2 + 5\,S + 16)\,Y(s) = 9\,X(s)$

$\frac{Y(S)}{X(S)} = \frac{9}{S^2 + 5\,S + 16}$

$\omega_n^2 = 16$

$\omega_n = 4$ rad / sec.

$2\,\zeta\omega_n = 5$

$\zeta = \frac{5}{2\,\omega_n} = 0.625$

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

$= 4\sqrt{1-(0.625)^2}$

$= 3.1225$

$t_d = \frac{1+0.7\,\zeta}{\omega_n} = 0.3593$ sec.

$t_r = \frac{\pi-\theta}{\omega_d}$

$\emptyset = \tan^{-1}\frac{\zeta}{\sqrt{1-\zeta^2}} = 0.6751$ rad

tr $= \frac{3.14-0.6751}{3.1225} = 0.719$ sec

$t_p = \frac{\pi}{\omega_d} = 1.005$ sec

$\% \text{ Mp} = 100 \times e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = 8.09\%$

$t_s = \frac{4}{\zeta\,\omega_n} = 1.6$ sec

Expression for output

$$\frac{Y(S)}{X(S)} = \frac{9}{16}\left[\frac{16}{S^2+5\,S+16}\right]$$

$$C(t) = \frac{9}{16}\left[1-\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\ Sin\,(\omega_d t+\phi)\right]$$

$$c(t) = \frac{9}{16}\,[1-1.2810\,e^{-2.5\,t}\,Sin\,(3.1225t+0.675]$$

Ex. Find K to limit the error of a system for input $1 + 8\,t + \frac{18}{2}t^2$ to 0.8 having

$$G(s)\,H(s) = \frac{K}{S^2\,(S+1)(S+4)} \qquad r(t) = 1 + 8\,t + \frac{18}{2}t^2$$

Solution : $R(s) = \frac{1}{S} + \frac{8}{S^2} + \frac{18}{S^3} = \frac{S^2+8\,S+18}{S^3}$

$$e_{ss} = \lim_{S\to 0}\frac{S\,R\,(S)}{1+G(S)\,H(S)}$$

$$= \lim_{S\to 0}\frac{S\left(\frac{S^2+8\,S+18}{S^3}\right)}{1+\frac{k}{S^2\,(S+1)\,(S+4)}}$$

$$= \lim_{S\to 0}\frac{S^2+8\,S+18}{S^2+\frac{K}{(S+1)\,(S+4)}}$$

$$= \frac{18}{\left(\frac{K}{4}\right)} = \frac{72}{k}$$

$$\frac{72}{K} = 0.8$$

$$K = \frac{72}{0.8} = 90$$

Ex. A system has G(s) = $\frac{K}{S\,(1+ST)}$ with unity feedback when K and T are const. Determine the factor by which K should be multiplied to reduce the overshoot from 85% to 35%.

G(s) = $\frac{K}{S\,(1+ST)}$ and H(s) = 1

$$\frac{C(s)}{H(s)} = \frac{G(s)}{1+G(s)\,H(s)}$$

$$= \frac{\frac{K}{S(1+ST)}}{1+\frac{K}{S\,(1+ST)}}$$

$$= \frac{K}{S^2\,T+S+K} = \frac{\left(\frac{K}{T}\right)}{S^2+\frac{1}{T}S+\frac{K}{T}}$$

$\omega_n{}^2 = \frac{K}{T}$, $\omega_n = \sqrt{\frac{K}{T}}$ rad/sec.

$2\,\zeta\,\omega_n = \frac{1}{T} = \frac{1}{2\omega_n T}$

$$\zeta = \frac{1}{2\left(\sqrt{\frac{K}{T}}\right)T} = \frac{1}{2\sqrt{KT}}$$

Ex. Find the unity feedback system having G(s) = $\frac{K}{S\,(ST+2)}$, find the following.

(a) The factor by which the gain K should be multiplied to increase the damping ratio from 0.15 to 0.6.

(b) The factor by which the time const. T should be multiplied to reduce the damping ratio from 0.8 to 0.4.

Solution : G(s) = $\frac{K}{S(ST+2)}$ H(S) = 1

$$\frac{C(s)}{H(s)} = \frac{G(s)}{1+G(s)\,H(s)} = \frac{K}{S^2T+2\,S+K}$$

$$= \frac{\frac{K}{T}}{S^2+\frac{2}{T}S+\frac{K}{T}}$$

$\omega_n{}^2 = \frac{K}{T}$, $\omega_n = \sqrt{\frac{K}{T}}$

$2\,\zeta\,\omega_n = \frac{2}{T}$

$$\zeta = \frac{2}{T}\,\frac{1}{2\omega_n} = \frac{1}{T\omega_n}$$

$$\zeta = \frac{1}{T\sqrt{\frac{K}{T}}} = \frac{1}{\sqrt{T\,.\sqrt{T}\,\frac{\sqrt{K}}{\sqrt{T}}}} = \frac{1}{\sqrt{KT}}$$

Let $\zeta = \zeta_1$ for Mp = 85% or 0.85

$$0.85 = e^{-\frac{\zeta_1\,\pi}{\sqrt{1-\zeta_1^2}}}$$

$$\ln(0.85) = \ln\left(e^{-\frac{\zeta_1\,\pi}{\sqrt{1-\zeta_1^2}}}\right)$$

$$\sqrt{1-\zeta_1^2} = (19.32)\ \zeta_1$$

$$(1-\zeta_1^2) = (373.264)\ \zeta_1^2$$

$$\zeta_1 = 0.01517 \quad - \quad (1)$$

Let $\zeta = \zeta_2$ for Mp = 35% or 0.35

$$0.35 = e^{-\frac{\zeta_2\,\pi}{\sqrt{1-\zeta_2^2}}}$$

$$\sqrt{1-\zeta_2^2} = 2.99\,\zeta_2$$

$$\zeta_2 = 0.317 \quad - \quad (2)$$

$$\frac{\zeta_1}{\zeta_2} = \frac{0.0517}{0.317}$$

$$\frac{\frac{1}{2\sqrt{K_1T}}}{\frac{1}{2\sqrt{K_2T}}} = \frac{0.0517}{0.317} \quad \left\{\therefore\ \zeta = \frac{1}{2\sqrt{KT}}\right\}$$

$$K_2 = 0.02659\ K_1$$

$$\therefore\ \zeta = \frac{1}{\sqrt{KT}}$$

Let $\zeta_1 = \frac{1}{\sqrt{K_1 T}}$

$\zeta_2 = \frac{1}{\sqrt{K_2 T}}$ for $\zeta = 0.15$ and $\zeta = 0.6$ respectively.

$$\frac{\zeta_1}{\zeta_2} = \sqrt{\frac{K_2 T}{K_1 T}} = \frac{0.15}{0.60} = \frac{1}{4}$$

$$\frac{K_2}{K_1} = \frac{1}{16}$$

$$K_2 = \frac{K_1}{16}$$

The gain must be multiplied by factor $\frac{1}{16}$ to increase the damping ratio from 0.15 to 0.6.

Let $\zeta_1 = \frac{1}{\sqrt{T_1 K}}$

$\zeta_2 = \frac{1}{\sqrt{T_2 K}}$ for $\zeta = 0.8$ and $\zeta = 0.4$ respectively.

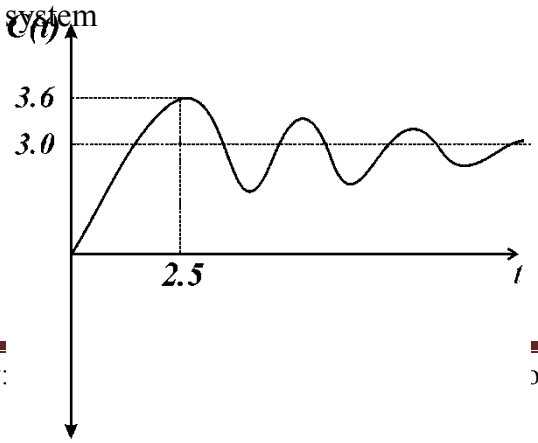
$$\frac{\zeta_1}{\zeta_2} = \sqrt{\frac{T_2 K}{T_1 k}} = \frac{0.8}{0.4} = 2$$

$$\frac{T_2}{T_1} = 4$$

$$T_2 = 4\,T_1$$

The time const. T must be multiplied by actor 4 to reduce the damping ratio from 0.8 to 0.4.

Ex. The figure shows step response of a second order system having unity feedback when input is $3u(t)$. Determine the open loop and closed loop transfer function of the system



Solution :

From figure tp = 2.5 sec.

Overshoot = 3.6 – 3.0 = 0.6 for final output 3

Mp $= \frac{3.6-3.0}{3.0} = \frac{0.6}{3} = 0.2$

% Mp $= \frac{0.6}{3}$ x 100 = 20%

% Mp $= e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$ x 100

0.2 $= e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$ x 100

- 1.61 = - $\frac{\pi\,\zeta}{\sqrt{1-\zeta^2}}$

ζ = 0.456

tp $= \frac{\pi\,\zeta}{\omega_n\sqrt{1-\zeta^2}}$

0.2 = ω_n = 1.414 rad/sec.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{(1.414)^2}{s^2 + 2(0.456)(1.414)S + 1.414)^2}$$

$$= \frac{(1.414)^2}{s^2 + 1.289S + (1.414)^2}$$

Let H(s) = 1

$$G(s) = \frac{1.999}{s^2 + 1.2895}$$

Disadvantages of static error coefficient method :

1) They do not give any information on the steady state error when the inputs are other than the standard test inputs (step, ramp and parabolic).
2) This method does not provide precise value of error. In most of the cases the error found in either zero or infinite.
3) It does not provide variation of error with respect to time which may be needed for design purpose.
4) The method is applicable to stable system.

Ex. Determine the step, ramp and parabolic error constants of the following feedback control systems.

(i) $G(s) = \frac{1200}{(1+0.25)(1+115)}$, $H(s) = 1$

Ans: Kp = 1200, Kv = 0, Ka = 0

(ii) $G(s) = \frac{120}{S(S^2+10S+120)}$, $H(s) = 1$

Ans: Kp = ∞, Kv = 1, Ka = 0

(iii) $G(s) = \frac{110}{S(1+0.25)(1+12s)}$, $H(s) = 1$

Ans: Kp = ∞, Kv = 110, Ka = 0

(iv) $G(s) = \frac{120}{S(S^2+105+110)}$, H(s)

Ans: Kp = ∞, Kv = ∞, Ka = 1

(v) $G(s) = \frac{1100}{S(S+10)(S+110)}$, $H(s) = 1$

Ans: Kp = ∞, Kv = ∞, Ka = 0

(vi) $G(s) = \frac{10(1+4S)(1+6S)}{S^2(S^2+2S+1)}$, $H(s) = 1$

Ans: Kp = ∞, Kv = ∞, Ka = 10

Objective type Questions:

1. Find the type for the system given by the following transfer function
   $C(s) / R(s) = (10(s + 5)^2) / (s^3(s + 2)(s + 10))$.
   Ans: 3
2. What is the value of K for a unity feedback (negative) system with open loop transfer function $G(s) = K / (S(S+1)(S+3))$ to get $e_{ss} = 0.2$ when a unit ramp input is applied?
   Ans: 15
3. For a 2nd order system, if the peak time is 2 times the rise time ($t_p = 2t_r$), then
   ξ = _____.

   Ans: 0
4. The step error co-efficient of a system $G(s) = \frac{1}{(s+6)(s+1)}$ with unify feedback is___
   Ans: 1/6
5. Consider the feedback control system with loop-transfer function
   $G(s)\,H(s) = \frac{K\,(1+0.5s)}{s\,(1+s)(1+2s)}$. The type of closed loop system is_____.

   Ans: One
6. Consider the unity feedback control system with open-loop transfer function $G(s) = \frac{K}{s\,(s+1)}$. The steady state error of the system due a unit step input is _____.
   Ans: K
7. For a second-order system with a closed-loop transfer function $T(s) = \frac{9}{s^2+4s+9}$ the settling time for 2-percent band, in seconds, is _____.
   Ans: 4
8. Consider a system with transfer function $G(s) = \frac{s+6}{Ks^2+s+6}$. Its damping ration will be 0.5 when the value of K is _____.
   Ans: 3
9. The open loop gain for a unity feedback system is $G(s) = \frac{12}{s(s+12)}$. The steady state velocity error of the system is _____.
   Ans: 1
10. Which of the systems is generally preferred?
    Ans: underdamped
11. If the characteristic equation of a closed loop system is $s^2+2s+2=0$,then the system is _____.

Ans: Critically damped

12. The natural frequency of an undamped second order system is 40 rad/s. If the system is damped with a damping ratio of 0.3, the damped natural frequency in rad/s is_____

    Ans: 38.15 r/sec

13. A critically damped continuous time second order system when sampled will have _____.

    Ans: Two simple poles on negative real axis

14. The steady state error of a stable type 0 unity feedback system for a unit step function is_____.

    Ans:1/1+kp

15. A unity feedback control system has the open-loop transfer function $G(s) = \frac{4(1+2s)}{s^2(s+2)}$ .If the input to the system is a unit ramp, the steady –state error will be_____.

    Ans: Zero

16. The response of an LCR circuit to a step input is _____.

    Ans: Oscillatory if the poles are imaginary

17. The transfer function $\mathrm{T}(s) = \frac{100}{(s+1)(s+100)}$ for a unit step input to the system the approximate settling time for 2-percent band in seconds, is _____.

    Ans: 4 sec

18. The transfer function $\mathrm{G}(s) = \frac{1}{(s+2)}$ is excited with 10u(t). The time at which the output reaches 99% of its steady state value is _____.

    Ans: 2.3 sec

19. A ramp input applied to a unity feedback system results in 5% steady state error. The type number and zero frequency gain of the system are respectively.

    Ans: 1 and 20

20. The unit step response of a system starting from rest is given by for $c(t) = 1 - e^{-2t}$ ($t \geq 0$) the transfer function of the system is _____.

    Ans: $\frac{2}{(s+2)}$